

5. Compressor Bearing Vibration Fitness-for-Service Review

To: Rotating Equipment Department, Operations **From:** M. Petrov, Rotating Equipment Engineer **Date:** 2026-06-30 **Subject:** Compressor C-101 — Bearing Vibration Fitness-for-Service Result (per SOP-RE-225 Clause 6.3)

Background

Per SOP-RE-225 Section 6, Clause 6.3, sustained vibration trend increases on critical rotating equipment exceeding the Clause 6.5 alarm-persistence threshold require a formal bearing fitness-for-service review before continued operation is authorized. C-101's thrust-bearing vibration trend exceeded this threshold for three consecutive monitoring periods.

Assessment Result

The evaluated sustained vibration limit for continued operation of C-101's thrust bearing, based on OEM allowable limits and current operating speed, is **4.5 mm/s RMS**. This value supersedes the general alarm setpoint used prior to this review.

Any sustained reading above 4.5 mm/s RMS for more than one monitoring period shall be treated as a fitness-for-service exceedance: the unit shall be ramped to reduced load and shall not be returned to full rated speed until re-evaluated per SOP-RE-225 Clause 6.4.

Distribution

Filed to the Rotating Equipment Integrity File per SOP-RE-225 Clause 8.1. Retain for the operating life of the unit plus 10 years.